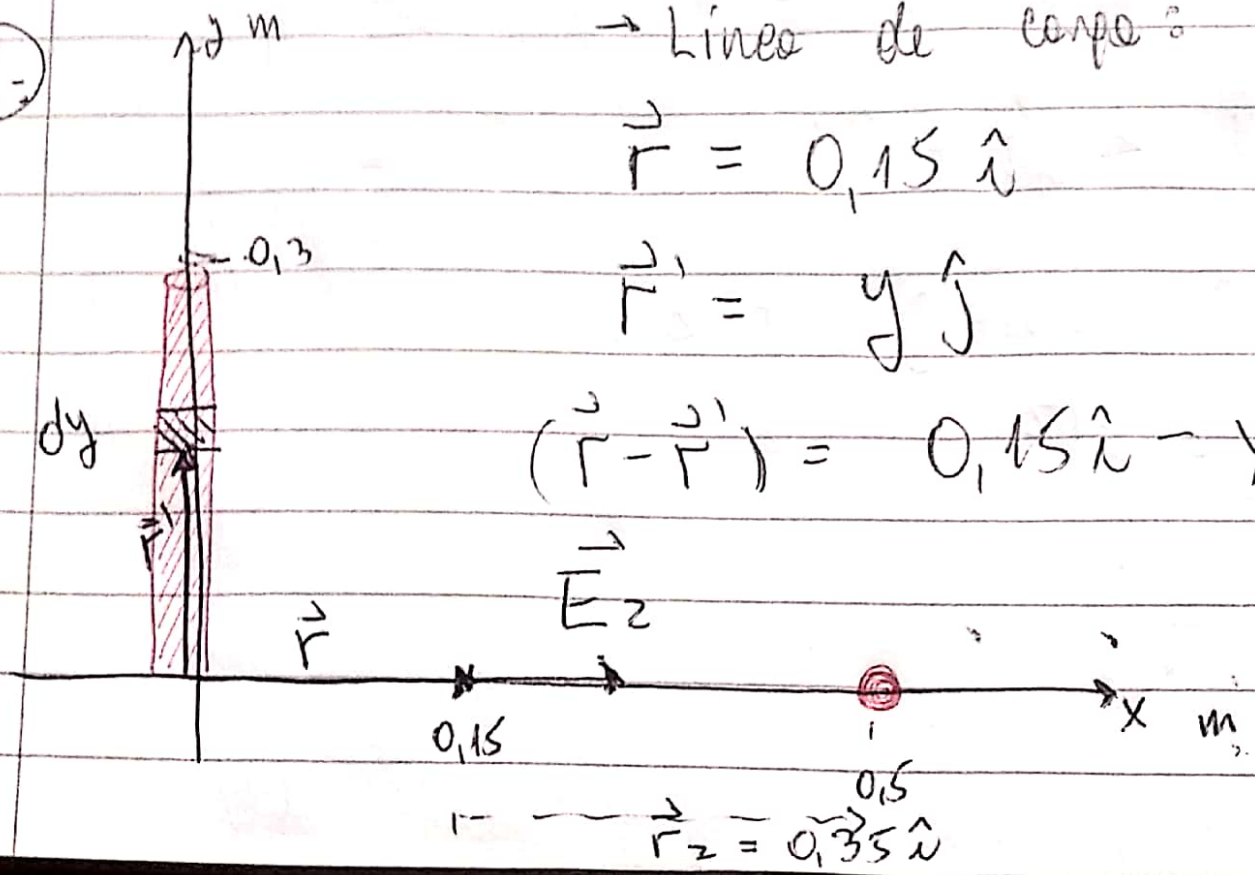
2.-



→ Linea de campo:

$$\vec{r} = 0{,}15\,\hat{\imath}$$

$$\vec{r}\,' = y\,\hat{\jmath}$$

$$(\vec{r} - \vec{r}\,') = 0{,}15\,\hat{\imath} - y\,\hat{\jmath}$$

Escaneado con CamScanner

$$0 \leq y \leq 0{,}3$$

$$\|\vec{r}-\vec{r}\,'\|^3 = (0{,}15^2 + y^2)^{3/2}$$

$$\frac{dq}{dy} = \lambda \qquad \Rightarrow d\vec{E} = K\frac{\lambda\, dy}{(0{,}15^2+y^2)^{3/2}}(0{,}15\,\hat{\imath} - y\,\hat{\jmath})$$

$$d\vec{E} = K\lambda\left\{\frac{0{,}15\,dy\,\hat{\imath}}{(0{,}15^2+y^2)^{3/2}} - \frac{y\,dy}{(0{,}15^2+y^2)^{3/2}}\,\hat{\jmath}\right\}$$

$$\vec{E} = K\lambda\left\{\int_0^{0{,}3}\frac{0{,}15}{(0{,}15^2+y^2)^{3/2}}\,dy\,\hat{\imath} - \int_0^{0{,}3}\frac{y\,dy}{(0{,}15^2+y^2)^{3/2}}\,\hat{\jmath}\right\}$$

$$= 9\cdot 10^9 \cdot 4\cdot 10^{-9}\cdot 0{,}15\int_0^{0{,}3}\frac{dy}{(0{,}0225+y^2)^{3/2}}\,\hat{\imath}$$

$$- 9\cdot 10^9 \cdot 4\cdot 10^{-9}\int_0^{0{,}3}\frac{y\,dy}{(0{,}0225+y^2)^{3/2}}\,\hat{\jmath}$$

Escaneado con CamScanner

Así:

$$\vec{E} = 5{,}4 \left. \frac{y}{0{,}0225\,(y^2+0{,}0225)^{1/2}} \right|_0^{0{,}3} \hat{\imath}$$

$$-36 \left. \frac{-1}{\sqrt{0{,}0225+y^2}} \right|_0^{0{,}3} \hat{\jmath}$$

$$= 5{,}4\,(39{,}752 - 0)\,\hat{\imath} - 36\,(-2{,}981+6{,}667)\,\hat{\jmath}$$

$$= 214{,}661\,\hat{\imath} - 132{,}696\,\hat{\jmath} \ [N/C]$$

El campo de la carga puntual

$$\vec{E}_2 = 9\cdot 10^9 \frac{5\cdot 10^{-3}}{(0{,}35)^3}\, 0{,}35\,\hat{\imath}$$

$$= 3{,}673\cdot 10^8\,\hat{\imath} \ [N/C].$$

$\Rightarrow$ El campo total

$$\vec{E}_T = \vec{E}_2 + \vec{E}$$

$$= 3{,}673\cdot 10^8\,\hat{\imath} - 132{,}696\,\hat{\jmath} \ [N/C].$$

Escaneado con CamScanner